LLM-Based YouTube Video Summarizer and Question Answering System

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Team Statement

We are going to work on this project as a team. Both team members will contribute to planning, development, testing, documentation, and final presentation of the YouTube Video Summarizer and Question Answering system.

Project Overview

This project aims to develop an AI-based web application that summarizes YouTube videos and answers user questions related to the video content using Large Language Models (LLMs). The user will paste a YouTube video link, and the system will extract the video transcript, generate a concise summary, display key points, and allow users to ask questions based on the video. This system will help users quickly understand long educational videos, lectures, tutorials, and presentations without watching the full video.

Problem Statement

Many YouTube videos are long and time-consuming, especially educational lectures, tutorials, and technical content. Users often need a faster way to understand the important concepts and ask specific questions without watching the complete video. This project solves the problem by automatically converting video transcripts into concise summaries and enabling an AI-powered question answering system based on the video content.

Objectives

The main objective is to build a practical AI-powered learning assistant. The system will extract transcripts from YouTube videos, summarize the content using an LLM, generate key points, and provide an interactive Question Answering feature where users can ask any question related to the video transcript.

Proposed Features

• Paste YouTube video URL
• Automatic transcript extraction
• AI-generated summary
• Important key points extraction
• Interactive Question Answering based on video content
• Optional quiz or study questions for revision
• Simple and user-friendly interface

Tools and Technologies

The project will use Python as the main programming language and Streamlit for the web interface. The transcript will be extracted using the YouTube Transcript API. For summarization and question answering, LLMs such as Google Gemini API, OpenAI API, or Hugging Face models like T5, BART, or Mistral can be used.

Methodology

First, the user enters a YouTube video link. The system extracts the video transcript using the YouTube Transcript API. The transcript is cleaned and divided into manageable chunks if required. The LLM then generates a concise summary and key points. The transcript data is stored temporarily so users can ask questions related to the video. The Question Answering module retrieves relevant transcript context and generates accurate answers based on the video content.

Expected Outcome

The final system will allow users to summarize YouTube videos quickly and ask questions related to the video content. It will reduce study time, improve understanding, and provide smart revision support for educational and learning purposes.

Conclusion

This project is simple, practical, and highly useful for students and learners. It demonstrates how Large Language Models can be applied to real-world educational problems by transforming long video content into meaningful summaries and interactive AI-powered learning assistance.